SQL Handbook (1–50 Commands with Before & After Tables - Exact Model)

1 SELECT

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
```

```
SQL:
SELECT * FROM students;
```

```
After:
| Ravi | 20 | Chennai |
| Kumar | 21 | Madurai |
| Anu | 19 | Coimbatore |
```

2 SELECT Columns

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
```

```
SQL:
SELECT name FROM students;
```

```
After:
| Ravi |
| Kumar |
| Anu |
```

3 WHERE

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
```

```
SQL:
SELECT * FROM students WHERE age > 20;
```

```
After:
| Kumar | 21 | Madurai |
```

4 INSERT

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
```

```
SQL:
INSERT INTO students VALUES (4,'Raj',22,'Trichy');
```

```
After:
New row added
```

5 UPDATE

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
```

```
SQL:
UPDATE students SET age=21 WHERE id=1;
```

```
After:
Ravi age updated to 21
```

6 DELETE

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
```

```
SQL:
DELETE FROM students WHERE id=3;
```

```
After:
Anu removed
```

7 ORDER BY

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
```

```
SQL:
SELECT * FROM students ORDER BY age DESC;
```

```
After:
Kumar, Ravi, Anu
```

8 LIMIT

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
```

```
SQL:
SELECT * FROM students LIMIT 2;
```

```
After:
First 2 rows
```

9 DISTINCT

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
```

```
SQL:
SELECT DISTINCT city FROM students;
```

```
After:
```

Chennai, Madurai, Coimbatore

10 LIKE

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
```

```
SQL:
SELECT * FROM students WHERE name LIKE 'R%';
```

```
After:
Ravi
```

11 IN

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
```

```
SQL:
SELECT * FROM students WHERE city IN ('Chennai','Madurai');
```

```
After:
Ravi, Kumar
```

12 BETWEEN

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
```

```
SQL:
SELECT * FROM students WHERE age BETWEEN 19 AND 21;
```

```
After:
All rows
```

13 COUNT

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
```

```
SQL:
SELECT COUNT(*) FROM students;
```

```
After:
3
```

14 SUM

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
```

SQL:
SELECT SUM(age) FROM students;

After:
60

15 AVG

Before Table (students):
id	name	age	city
1	Ravi	20	Chennai
2	Kumar	21	Madurai
3	Anu	19	Coimbatore

SQL:
SELECT AVG(age) FROM students;

After:
20

16 MAX

Before Table (students):
id	name	age	city
1	Ravi	20	Chennai
2	Kumar	21	Madurai
3	Anu	19	Coimbatore

SQL:
SELECT MAX(age) FROM students;

After:
21

17 MIN

Before Table (students):
id	name	age	city
1	Ravi	20	Chennai
2	Kumar	21	Madurai
3	Anu	19	Coimbatore

SQL:
SELECT MIN(age) FROM students;

After:
19

18 GROUP BY

Before Table (students):
id	name	age	city
1	Ravi	20	Chennai
2	Kumar	21	Madurai
3	Anu	19	Coimbatore

SQL:
SELECT city,COUNT(*) FROM students GROUP BY city;

After:
Grouped count

19 HAVING

Before Table (students):
| id | name | age | city |

```
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
```

```
SQL:
SELECT city,COUNT(*) FROM students GROUP BY city HAVING COUNT(*)>0;
```

```
After:
All cities
```

20 JOIN

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
```

```
SQL:
SELECT * FROM students s JOIN courses c ON s.id=c.id;
```

```
After:
Joined rows
```

21 LEFT JOIN

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
```

```
SQL:
SELECT * FROM students s LEFT JOIN courses c ON s.id=c.id;
```

```
After:
All students
```

22 RIGHT JOIN

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
```

```
SQL:
SELECT * FROM students s RIGHT JOIN courses c ON s.id=c.id;
```

```
After:
All courses
```

23 UNION

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
```

```
SQL:
SELECT name FROM students UNION SELECT name FROM teachers;
```

```
After:
Unique names
```

24 UNION ALL

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
```

```
SQL:
SELECT name FROM students UNION ALL SELECT name FROM teachers;
```

```
After:
All names
```

25 CASE

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
```

```
SQL:
SELECT name,CASE WHEN age>20 THEN 'Senior' ELSE 'Junior' END FROM students;
```

```
After:
Category
```

26 SUBQUERY

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
```

```
SQL:
SELECT * FROM students WHERE age>(SELECT AVG(age) FROM students);
```

```
After:
Above avg
```

27 EXISTS

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
```

```
SQL:
SELECT * FROM students WHERE EXISTS(SELECT 1);
```

```
After:
All rows
```

28 NOT EXISTS

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
```

```
SQL:
SELECT * FROM students WHERE NOT EXISTS(SELECT 1 WHERE 1=0);
```

```
After:
```

All rows

29 CREATE TABLE

Before Table (students):
id	name	age	city
1	Ravi	20	Chennai
2	Kumar	21	Madurai
3	Anu	19	Coimbatore

SQL:
CREATE TABLE test(id INT);

After:
Table created

30 DROP TABLE

Before Table (students):
id	name	age	city
1	Ravi	20	Chennai
2	Kumar	21	Madurai
3	Anu	19	Coimbatore

SQL:
DROP TABLE test;

After:
Table deleted

31 ALTER ADD

Before Table (students):
id	name	age	city
1	Ravi	20	Chennai
2	Kumar	21	Madurai
3	Anu	19	Coimbatore

SQL:
ALTER TABLE students ADD email VARCHAR(100);

After:
Column added

32 ALTER MODIFY

Before Table (students):
id	name	age	city
1	Ravi	20	Chennai
2	Kumar	21	Madurai
3	Anu	19	Coimbatore

SQL:
ALTER TABLE students MODIFY name VARCHAR(100);

After:
Column changed

33 ALTER DROP

Before Table (students):
id	name	age	city
1	Ravi	20	Chennai
2	Kumar	21	Madurai
3	Anu	19	Coimbatore

```
SQL:
ALTER TABLE students DROP COLUMN email;
After:
Column removed
```

34 INDEX

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
SQL:
CREATE INDEX idx_name ON students(name);
After:
Index created
```

35 DROP INDEX

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
SQL:
DROP INDEX idx_name ON students;
After:
Index removed
```

36 VIEW

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
SQL:
CREATE VIEW v AS SELECT name FROM students;
After:
View created
```

37 SELECT VIEW

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
SQL:
SELECT * FROM v;
After:
View result
```

38 DROP VIEW

```
Before Table (students):
| id | name | age | city |
```

```
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
```

```
SQL:
DROP VIEW v;
```

```
After:
View deleted
```

39 TRANSACTION

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
```

```
SQL:
START TRANSACTION;
```

```
After:
Transaction started
```

40 COMMIT

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
```

```
SQL:
COMMIT;
```

```
After:
Changes saved
```

41 ROLLBACK

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
```

```
SQL:
ROLLBACK;
```

```
After:
Changes undone
```

42 LENGTH

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
```

```
SQL:
SELECT LENGTH(name) FROM students;
```

```
After:
Length values
```

43 UPPER

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
```

```
SQL:
SELECT UPPER(name) FROM students;
```

```
After:
Uppercase
```

44 LOWER

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
```

```
SQL:
SELECT LOWER(name) FROM students;
```

```
After:
Lowercase
```

45 CONCAT

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
```

```
SQL:
SELECT CONCAT(name,city) FROM students;
```

```
After:
Combined
```

46 NOW

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
```

```
SQL:
SELECT NOW();
```

```
After:
Current time
```

47 CURDATE

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
```

```
SQL:
SELECT CURDATE();
```

```
After:
```

```
Today date
```

48 ROUND

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
```

```
SQL:
SELECT ROUND(20.567,2);
```

```
After:
20.57
```

49 FLOOR

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
```

```
SQL:
SELECT FLOOR(20.9);
```

```
After:
20
```

50 CEIL

```
Before Table (students):
| id | name | age | city |
| 1 | Ravi | 20 | Chennai |
| 2 | Kumar | 21 | Madurai |
| 3 | Anu | 19 | Coimbatore |
```

```
SQL:
SELECT CEIL(20.1);
```

```
After:
21
```